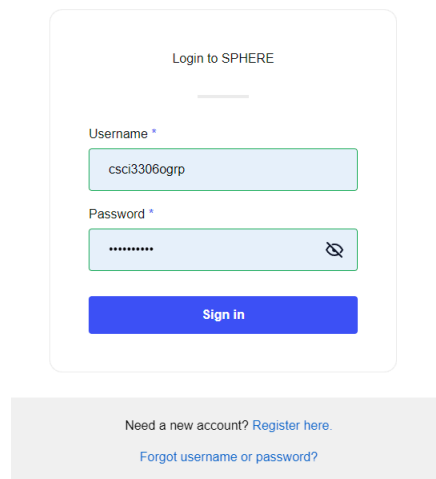


Lab 1

Stephanie Salgado

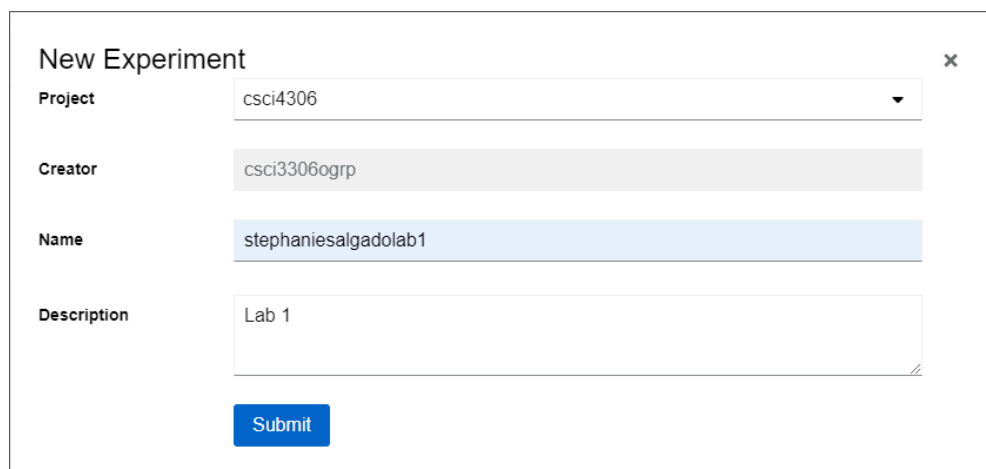
Accessing Deterlab (New Platform)



The screenshot shows a login form titled "Login to SPHERE". It contains two input fields: "Username" with the value "csci3306ogrp" and "Password" with masked characters "*****". A blue "Sign in" button is positioned below the password field. At the bottom of the form, there are two links: "Need a new account? Register here." and "Forgot username or password?".

I logged into the platform using the provided login information.

Creating an Experiment



The screenshot shows a "New Experiment" form with a close button (X) in the top right corner. The form has four fields: "Project" (a dropdown menu showing "csci4306"), "Creator" (a text field showing "csci3306ogrp"), "Name" (a text field showing "stephaniesalgadolab1"), and "Description" (a text area showing "Lab 1"). A blue "Submit" button is located at the bottom left of the form.

Navigated to "Experiments" on the side bar and clicked "New Experiment".

Viewing Experiment Details

Projects > csci4406 > Experiments > stephaniesalgadolab1

Experiment: csci4306.stephaniesalgadolab1

Details

Name	Project	Repository	Description	Creator
stephaniesalgadolab1	csci4306	https://git.sphre-testbed.net/csci4306/stephaniesalgadolab1		csci3306gdp

Models


No revisions found

Status

☐ ☒ ☐

Name	Description	Status	Creation	Last Updated
✓ Experiment: csci4306.stephaniesalgadolab1	csci4306stephaniesalgadolab1	Success	8/29/2024, 10:33:42 AM	8/29/2024, 10:33:43 AM
>  gitgit	Process experiment repositories	Success	8/29/2024, 10:33:42 AM	8/29/2024, 10:33:43 AM
>  modelmodel	Compile experiments	Success	8/29/2024, 10:33:42 AM	8/29/2024, 10:33:43 AM

Model Editor



Start editing

Drag and drop a file or upload one.

Browse

Start from scratch

In “Model Editor”, I selected “Start from scratch” and added the example provided in the documentation.

```
from mergexp import *

# Create a network topology object. This network will automatically
# add IP addresses to all node interfaces and configure static routes
# between all experiment nodes.
net = Network('hello-world', addressing==ipv4, routing==static)

# Create three nodes.
a,b,c = [net.node(name) for name in ['a', 'b', 'c']]

# Create a link connecting the three nodes.
link = net.connect([a,b,c])

# Make this file a runnable experiment based on our three node topology.
experiment(net)
```

Compilation

✓ Compilation Successful

```
1 from mergexp import *
2
3 # Create a network topology object. This network will automatically
4 # add IP addresses to all node interfaces and configure static routes
5 # between all experiment nodes.
6 net = Network('hello-world', addressing==ipv4, routing==static)
7
8 # Create three nodes.
9 a,b,c = [net.node(name) for name in ['a', 'b', 'c']]
10
11 # Create a link connecting the three nodes.
12 link = net.connect([a,b,c])
13
14 # Make this file a runnable experiment based on our three node topology.
15 experiment(net)
```

Upon compiling successfully, we get the “Compiled Topology” pictured below.

Compiled Topology

Revision Details

ID: hello-world

Model View

- ☒ Nodes
- ☐ Nodes & LANs
- ☐ Nodes & Interfaces

Topology View

- ☒ 2D
- ☐ 3D

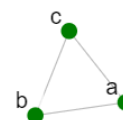
Topology Display

Node Size: 3

0 10

Font Scale: 20

0 96



Pushing Model to Experiment

Push Model to Experiment ×

Experiment *

stephaniesalgadolab1.csci4306

Branch

master

Tag

Push

✓ Model Pushed

Revision created in [stephaniesalgadolab1.csci4306](#): [5ae306b73f60f174b0b596e4f3ccde828818542c](#)

After that, I push the model to the experiment I had created. Once pushed, there is a “Model Pushed” success message.

Details

Name	Project	Repository
stephaniesalgadolab1	csci4306	https://git.sphere-testbed.net/csci4306/stephaniesalgadolab1

Models

Revision	Reservations	Compilation	Timestamp
5ae306b7		Succeeded	8/29/2024, 10:47:52 AM

Status

Name	Description	Status
Experiment: csci4306/stephaniesalgadolab1	csci4306/stephaniesalgadolab1	Success
> /git/git	Process experiment repositories	Success
> /model/model	Compile experiments	Success

When I inspect the details, this time there is a that shows that the compilation succeeded.

Realizing the Experiment

Reserve Revision



A Reservation is a request to exclusively reserve resources for an experiment. This is an asynchronous process. Check the reservations listing or the dashboard to see the status of your reservation. If the reservation fails, it is likely that your model is requesting resources that are not currently available. In that case, relinquish the reservation and try again at a later time.

Project

csci4306

Experiment

stephaniesalgadolab1

Reservation Name

helloworldssalgado

Submit

To realize the revision, I clicked on the revision hash then “Reserve Revision”.

Assessing Realization Status

Reservations



Reservation ↑	Experiment ↑	Project ↑
helloworldbradfield	johnbradfieldlab1	csci4306
helloworldcatalinal	catalinalemmonlab1	csci4306
helloworldgonzalez	andrewgonzalezlab1	csci4306
helloworldssalgado	stephaniesalgadolab1	csci4306

Under “Reservations”, I find the reservation I made under the name I gave it.

Projects > csci4306 > Experiments > stephaniesalgadolab1 > Reservations > helloworldssalgado

Reservation Node Details						
Name	Experiment Address	Interface Address	Kind	Resource	Image	Facility
a	10.0.1.124	172.20.0.11	VirtualMachine	PH01	ubuntu	ecobator
b	10.0.1.125	172.20.0.12	VirtualMachine	PH01	ubuntu	ecobator
c	10.0.1.123	172.20.0.13	VirtualMachine	PH01	ubuntu	ecobator

Reservation Details				
Owner	Created	Expires	Network	Attached Server
cscibiology	Thu, 26 Aug 2024 10:01:36 GMT	Thu, 01 Sep 2024 10:01:36 GMT	labnet001	811

Status				
Name	Description	Status	Creation	Last Update
Realization: helloworldssalgado.stephaniesalgadolab1.csci4306 Manage realizations	Manage Realization: helloworldssalgado.stephaniesalgadolab1.csci4306	Success	8/29/2024, 10:01:36 AM	8/29/2024, 11:01:36 AM
Realization: helloworldssalgado.stephaniesalgadolab1.csci4306 Manage realizations	Manage realizations	Success	8/29/2024, 11:01:36 AM	8/29/2024, 11:01:36 AM

Clicking on “helloworldssalgado” displays its details.

Materializing the Realization

Activation helloworldssalgado.stephaniesalgadolab1.csci4306 started.

To materialize, I go back to the previous screen and select the 3 dots next to my reservation, then I click “Activate”.

Activation

helloworldbradfield.johnbradfieldlab1.csci4306

helloworldcatalinal.catalinalemmonlab1.csci4306

helloworldgonzalez.andrewgonzalezlab1.csci4306

helloworldssalgado.stephaniesalgadolab1.csci4306

helloworldtello.jeremytello1.csci4306

Notice, when clicking “Activations” on the side bar, the activation appears.

Status

Details

Name	Description	Status
helloworldssalgado.stephaniesalgadolab1.csci4306	Materialization Management: helloworldssalgado.stephaniesalgadolab1.csci4306	Success
Portal Operations	All portal operations for helloworldssalgado.stephaniesalgadolab1.csci4306	Success
moddeter: helloworldssalgado.stephaniesalgadolab1.csci4306	Create materialization helloworldssalgado.stephaniesalgadolab1.csci4306	Success

Clicking on the activation reveals its status as success.

Attaching XDC

New Experiment Development Container ✕

Project

csci4306

Name

stephaniesalgadolab1

Type

shared

Submit

First, I created a new XDC.

Attach to Materialization ✕

XDC *

stephaniesalgadolab1.csci4306

Materialization *

helloworldssalgado.stephaniesalgadolab1.csci4306

Attach

 XDC stephaniesalgadolab1.csci4306 attached to helloworldssalgado.stephaniesalgadolab1.csci4306

Then, clicking on the 3 dots allows you to attach it to a materialization.

XDC Information CPU Limit

Name	Project	Type	Attached	URL	SSH Name	Creator	Mem...	CPU...	Image
stephaniesalgadolab1	csci4306	shared	helloworldssalgado.stephaniesalgadolab1.csci4306	Jupyter	stephaniesalgadolab1-csci4306	csci33060grp	2	2	xdc-base.v1.3.9

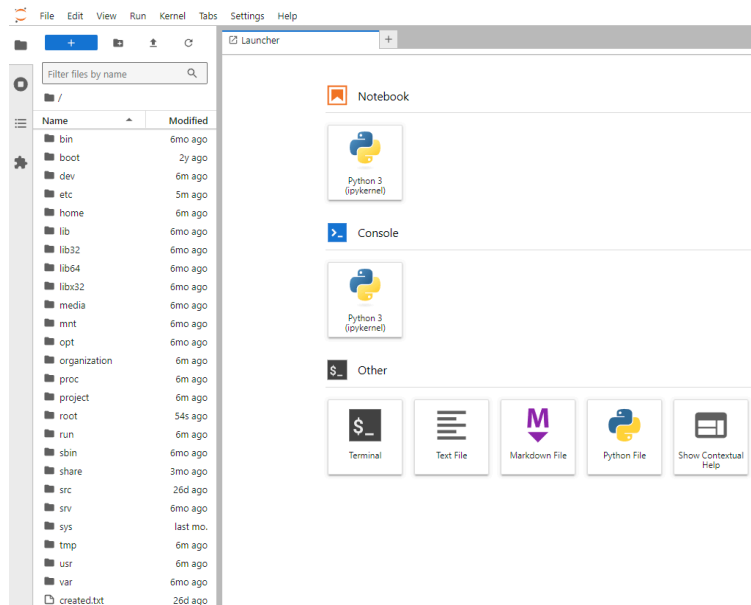
Status



Name	Description	Status	Creation	Last Updated
▼  XDC: stephaniesalgadolab1.csci4306	Manage XDC: stephaniesalgadolab1.csci4306	Success	8/29/2024, 11:11:54 AM	8/29/2024, 11:27:32 AM
>  /mergetfs/Creds	MergeFS Credentials Management	Success	8/29/2024, 11:25:27 AM	8/29/2024, 11:25:28 AM
>  /xdc/podinit	Initialize XDCs	Success	8/29/2024, 11:26:04 AM	8/29/2024, 11:26:07 AM
>  /wgsvc/WgXDCAttach	Manage wg xdc attachment	Success	8/29/2024, 11:26:02 AM	8/29/2024, 11:27:04 AM
>  /wgsvc/WgClient	Manage wireguard xdc connections	Success	8/29/2024, 11:27:03 AM	8/29/2024, 11:27:04 AM
>  /wgsvc/WgIf	Manage wireguard interfaces	Success	8/29/2024, 11:11:54 AM	8/29/2024, 11:27:04 AM
>  /credmgr/XDCCreds	Host credentials management	Success	8/29/2024, 11:25:26 AM	8/29/2024, 11:27:06 AM
>  /mergetfs/XcdFS	MergeFS XDC Management	Success	8/29/2024, 11:25:26 AM	8/29/2024, 11:27:04 AM
>  /xdc/Xdcs	Manage XDCs	Success	8/29/2024, 11:25:26 AM	8/29/2024, 11:27:09 AM

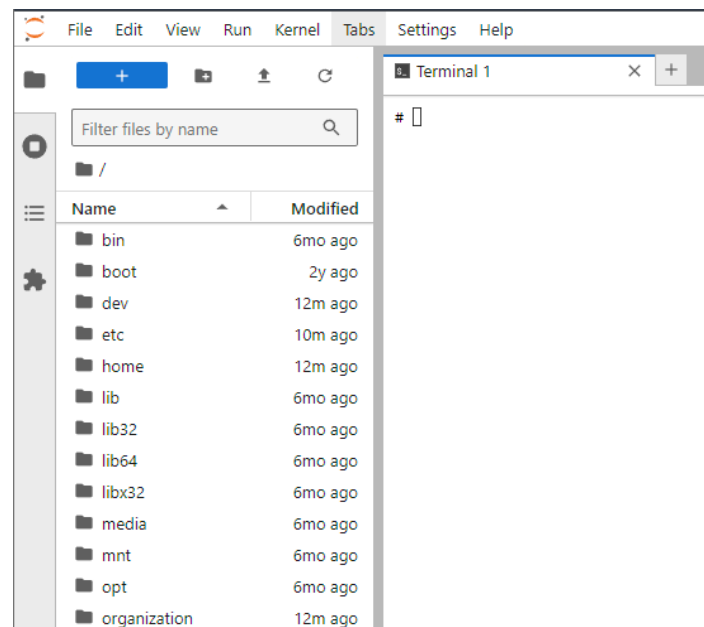
Clicking on it reveals its details.

Starting Instance

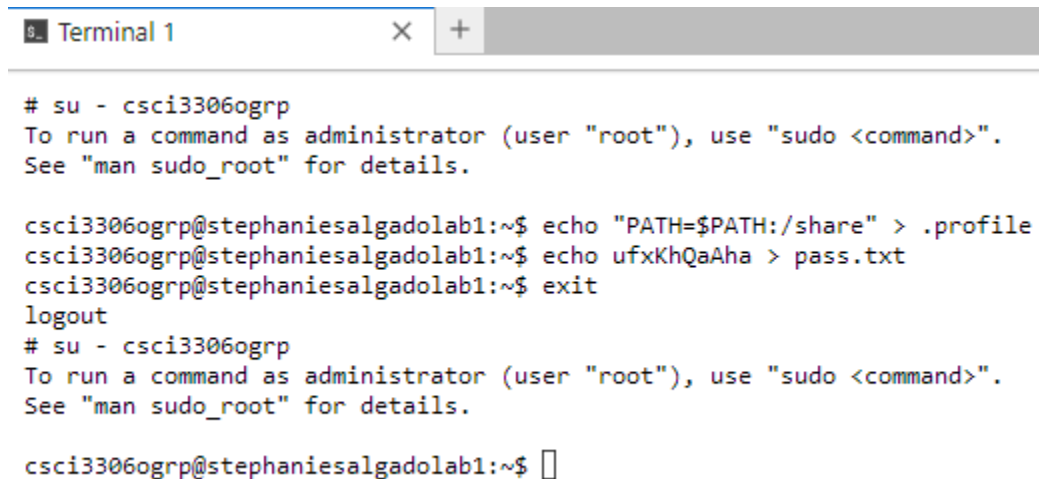


I go back to the list of XDCs and click Jupyter on my XDC to start my instance.

Terminal



I launch a terminal within the XDC.



```
# su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab1:~$ echo "PATH=$PATH:/share" > .profile
csci3306ogrp@stephaniesalgadolab1:~$ echo ufxKhQaAha > pass.txt
csci3306ogrp@stephaniesalgadolab1:~$ exit
logout
# su - csci3306ogrp
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

csci3306ogrp@stephaniesalgadolab1:~$
```

Since it's my first time accessing, I run the following commands.

Summary

In this lab, I accessed Deterlab and familiarized myself with the platform. I created an experiment, then created a model. After that I compiled and pushed the model to the experiment. Then I realized the experiment. Once that was done, I materialized the realization and attached it to my XDC. Lastly I started an instance and launched a terminal session on the XDC. Overall, this lab was a helpful introduction to the platform.